

INORGANIC ANALYSIS

I. PRELIMINARY TEST:

Color: Note down the color of the given salt.	Blue or Bluish green Greenish Light green Dark brown Pink Light pink or Flesh color White	Maybe Cu^{2+} or Ni^{2+} Maybe Ni^{2+} Maybe Fe^{2+} Maybe Fe^{3+} Maybe Co^{2+} Maybe Mn^{2+} May be $\text{Cu}^{2+}, \text{Co}^{2+}, \text{Ni}^{2+}, \text{Fe}^{3+}, \text{Mn}^{2+}$ etc are absent
Odor: Noted odor of the given salt	Smell of vinegar Smell of ammonia	May be acetate May be ammonium
Flame Test: Prepare a paste of the mixture with Con.HCl , in a clean watch glass. Place a small amount of this paste on the platinum wire loop and introduce it into the flame. Note the color imparted to the flame	Brick -red Crimson -red Grassy-green Or Apple green Bright bluish-green Green flashes Dull bluish - white	Ca^{2+} Sr^{2+} Ba^{2+} Cu^{2+} Zn^{2+} or Mn^{2+} Pb^{2+}
Ash test: Salt + ConHNO₃ + few drops of Cobalt Nitrate , boil the solution, dip a piece of filter paper and burn it. Note the color of ash	Green Pink Blue Black	Zn^{2+} Mg^{2+} Al^{3+} Cobalt

II. ANALYSIS OF ANIONS:

PROCEDURE	OBSERVATION	INFERENCE
1. To a little amount of salt add 2ml of dil.HCl	Brisk effervescence of a colorless gas.	CO₃²⁻ may be present
2. To the given substance add Con.H₂SO₄ The above solution is heated	Colourless gas with Smell of vinegar Reddish brown vapors. Colorless gas with pungent smell which gives dense white fumes with a glass rod dipped in NH₄OH Violet vapors	Acetate may be present Br ⁻ ion may be present Cl ⁻ ion may be present I ⁻ may be present
3. To the given substance add Con.H₂SO₄ . Heat and add paper ball	Dense brown fumes	NO₃⁻ may be present
4. To the salt solution, add a few drops of BaCl₂ solution	White precipitate which is insoluble in Dil.HCl	SO₄²⁻ may be present
5. To the salt solution add ammonium molybdate (NH₄)₂MoO₄ , Few	Yellow ppt	Phosphate ion may be present

drops of Con.HNO₃ heat		
6. O.S + dil.HNO₃ boil and add excess of NH₄OH . Boil off the excess NH₄OH and CaCl₂ solution	White ppt	Oxalate ion may be present

III. CONFIRMATORY TEST FOR ANIONS :

PROCEDURE	OBSERVATION	INFERENCE
<u>CARBONATE ION</u> Dil.H₂SO₄ test: To a little amount of the salt, add dil.H₂SO₄ Pass the gas evolved through lime water. 2. MgSO₄ test :- A little of the salt solution is treated with few drops of MgSO₄ solution	Brown effervescence and evolution of colorless odorless gases. Lime water turns milky White precipitate	The gas evolved is CO₂, CO₂ reacts with the lime water to form white precipitate of CaCO₃ which is responsible for the milkiness. Presence of Carbonate ion confirmed
<u>ACETATE ION</u> Ferric Chloride Test: Take the salt solution, add	Reddish coloration	Presence of Acetate is confirmed

neutral Ferric chloride solution Ester Test: Take a small quantity of the salt in a test tube. Add Con.H₂SO₄ (2ml) and heat. Now ethyl alcohol (1ml). Shake. Pour the contents of the tube in a beaker full of water. Stir.	Pleasant fruity smell of ester	
<u>CHLORIDE ION</u> 1.Silver nitrate test: Acidify a portion of aqueous solution with dil.HNO₃. Boil for some time, cool and add AgNO₃ solution 2.Chromyl chloride test: Mix a small quantity of the salt with a small amount of powdered Potassium dichromate and add Con.H₂SO₄. Heat the test tube and pass the red vapors evolved into a test tube containing NaOH solution. To the yellow solution thus obtained, Lead acetate solution. 3. Manganese dioxide test: Heat a pinch of the salt with a small quantity of MnO₂ and Con.H₂SO₄.	A white curdy precipitate is formed which is soluble in Ammonium hydroxide A yellow precipitate is formed. Evolution of greenish yellow gas having a pungent irritating smell	Presence of Cl⁻ ion is confirmed.

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<u>BROMIDE ION</u> Silver nitrate test: Acidify a portion of aqueous with dil. HNO₃. Boil, cool and add AgNO₃ solution Manganese dioxide test: Heat a small quantity of salt with solid MnO₂ and add Con.H₂SO₄ Layer test : To the original salt solution add 1ml of CHCl₃/CCl₄ and 2ml of Chlorine water and shake well.	A light-yellow precipitate is formed which is partially soluble in NH₄OH . Evolution of yellow brown vapor of bromine which turn starch paper yellow Orange or brown layer	Presence of Br⁻ ion is confirmed.
<u>IODIDE ION</u> Silver nitrate test: Acidify a portion of aqueous with dil.HNO₃. Boil, cool and add AgNO₃ solution Manganese dioxide test: Heat a small quantity of the salt with solid MnO₄ and Con.H₂SO₄ Layer test: To the original salt solution add 1ml of CHCl₃/CCl₄ and add 2ml of the chlorine water and shake well	A yellow precipitate is formed which is insoluble in NH₄OH . Evolution of violet vapors of Iodine which turn starch paper blue. Pink or violet layer	Presence of I⁻ ion is confirmed.

NITRATE ION

Copper chip test: Heat a small quantity of the original salt with concentrated H_2SO_4 and a few copper chips

Brown Ring test: Add a small quantity of freshly prepared solution of **ferrous sulphate** to a part of aqueous solution and then pour Con. H_2SO_4 slowly along the sides of the test tube.

SULPHATE ION

Lead acetate test: To a part of aqueous solution of the salt add Acetic acid and lead acetate solution

Dark brown fumes of **Nitrogen Dioxide** are evolved.

A dark brown ring is formed at the junction of the layers of the acid and the solution.

A white precipitate insoluble in dil. HCl , but soluble in excess of hot ammonium acetate solution.

Presence of NO_3^- confirmed.

Presence of NO_3^- confirmed.

Presence of sulphate ion is confirmed.

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IV. ANALYSIS OF CATIONS:

The remaining tests must be performed with a solution of the compound. If the compound is insoluble in water, dissolve it in dil.HCl

PROCEDURE	OBSERVATION	INFERENCE
GROUP 0: 1.To a small amount of solid salt taken in a test tube, add solution of sodium hydroxide and heat.	Smell of ammonia	Ammonium ion may be present
GROUP I : O.S + dil.HCl	White precipitate	1 st group cations are present.May be Pb²⁺
GROUP II : O.S +dil.HCl , warm.Pass H₂S gas through solution.	Black ppt.	2 nd group cations (Cu²⁺) May be present
GROUP III : O.S + NH₄Cl , shake well and add excess NH₄OH	White gelatinous ppt.	Group III cations (Al³⁺) May be present
GROUP IV : Add H₂S to the above	Black precipitate	Ni²⁺/Co²⁺

solution	Flesh colored ppt Dirty white ppt	Mn²⁺ Zn²⁺
GROUP V : O.S + NH₄Cl + NH₄OH add excess ammonium carbonate solution	White ppt	Group V May be present
GROUP VI : O.S + solid NH₄Cl + Sodium hydrogen phosphate solution and scratch the inner sides of the test tube with a glass rod.	A white precipitate	Mg²⁺ may be present

CONFIRMATORY TESTS FOR CATIONS

C.T for Test for NH₄⁺ i) Boil salt + NaOH solution. Expose a glass rod dipped in HCl to the gas ii) NESSLER'S REAGENT TEST: O.S + NESSLER'S reagent+ NaOH solution	Pungent smelling gas is evolved. Dense white fumes are observed A brown precipitate is obtained A yellow ppt is formed.	NH₄⁺ is confirmed. Presence of Lead ion is confirmed.
C.T for Lead ions i) O.S + Potassium chromate solution ii) O.S + KI solution.)Yellow ppt. + water, boil&cool.	Yellow ppt is formed. Precipitate dissolves in hot water and reappears as golden yellow spangles on cooling.	

<u>CT for Aluminum ion</u> 1. Lake test: The white gelatinous precipitate is dissolved in dil.HCl. Added two drops of blue litmus solution and added few drops of ammonium hydroxide till blue color develop. 2. Ash test .	Blue precipitate floating in colorless solution blue ash	Presence of Al^{3+} confirmed Presence of Al^{3+} confirmed
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GROUP IV

If black ppt,

a) **O.S + KNO₂ + Acetic acid**

Blue or yellow ppt

Presence of Cobalt confirmed.

b) **O.S + NH₄OH**, add few

Bright red coloration

Presence of Nickel confirmed.

drops of **dimethyl glyoxime**

Black ppt

Presence of Nickel confirmed.

c) **O.S + NaOH**

solution (in excess) – Green precipitate is formed-Add

Bromine water

to the above ppt

White precipitate turns black or brown

Presence of Mn²⁺ confirmed.

If flesh colored ppt,

O.S + NaOH solution

– A white precipitate is formed

-Add **Bromine water** to white precipitate

Pink colored solution is formed.

Presence of Mn²⁺ confirmed.

Lead peroxide test: To black ppt, obtained in above test (Sodium hydroxide- Br₂ test), add **con.HNO₃** and lead peroxide. Boil, cool and allow to settle.

White ppt which is soluble in excess NaOH

Presence of Zn²⁺ confirmed.

If white ppt,

1.O.S + NaOH solution.

White or bluish white ppt

Green ash

Presence of Zn²⁺ confirmed

**2. Potassium
ferrocyanide test:
O.S + K₄Fe(CN)₆**

3. Ash test

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GROUP V:

Dissolve the ppt in hot **dil.Acetic acid** and divide it into 3 parts.

C.T for Ba²⁺ :

i)Potassium chromate

test: One part of the solution + few drops of **potassium chromate** solution

ii)Flame test

Make a paste of the original salt in **con.HCl** on a watch glass. Take the salt paste on the loop of the **Pt** wire and introduce it into the flame. Note the color imparted to the flame.

C.T for Sr²⁺ :

i)Ammonium sulphate

test:

Second part of the solution + **Ammonium sulphate** solution, warm

ii)Flame test :

C.T for Ca²⁺ :

i)Ammonium Oxalate

test:

Third part of the solution +

Ammonium Oxalate

solution + few drops of **Ammonium hydroxide**, Scratch the inner sides of the test tube. **ii)Flame test**

A yellow ppt is formed.

Ba²⁺ is confirmed.

Apple green color

Ba²⁺ is confirmed.

White precipitate is formed

Sr²⁺ is confirmed.

Crimson red colored flame

Sr²⁺ is confirmed.

A white precipitate is formed

Ca²⁺ is confirmed.

Brick red flame

Ca²⁺ is confirmed.

GROUP VI :

1.O.S + solid **NH₄Cl** + **NH₄OH** solution in slight excess. Add **Ammonium phosphate** solution and scratch the inner sides of the test tube with a glass rod.

2. Ash test

A white precipitate is formed.

Pink colored ash

Mg²⁺ is confirmed .

Mg²⁺ is confirmed

